CIRC HQ STUDENT HANDOUT | GRADE 5

KidWind 1: Make the wind do work

Goal: I can build blades that turn wind into work and measure how far they lift a load.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 10, grade 5 path; Playbook B meeting 11, grade 5 path

Ask | Plan | **Build** | **Test** | Improve

You are done when

Say how many centimeters your blades lifted the cup. Name one thing you might change.

You will use

- One teacher-run test station, built and piloted first from parts on hand
- One guarded fan at one fixed setting, with tape marks and a ruler
- A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub
- String on a fixed spool, one small cup and the same load pieces every trial
- Cardstock or index cards, reusable blade supports, tape and student scissors
- A marked blade-angle example, timer, shallow catch tray, team folders and record sheets
- Printed practice data table for the no-fan paper path

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Sit with your team.
- Read the goal: I can make the wind do work.

2 Step 2: Tell your partner — 1 min

- Tell your partner one place you feel strong wind.

3 Step 3: Station safety — 5 min

- Watch the cup rise from the start line.
- Stay out of the spin path.
- Only the teacher turns the fan on or off.

4 Step 4: Build a rotor set — 14 min

- Cut three blades from cardstock.
- Tape each blade to a blade support.
- Set each blade at the same angle as the example.

5 Step 5: Test at the station — 8 min

- Your team gets one 10 second trial.
- Watch the cup rise.
- Record the centimeters lifted.

6 Step 6: Reset the station — 3 min

- Fan off first.
- Rotor sets go flat in your team folder.
- Cup and load pieces go back in the station tray.

7 Step 7: Show your result — 2 min

- Say how many centimeters your blades lifted the cup.
- Name one thing you might change.

What to turn in

Your team's centimeters lifted in 10 seconds, on your record sheet and the class chart.

No computer? Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Finished early? Next challenge

Cut a spare set of blades in a new shape and label what you would test next.

Stay safe

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.

No-fan paper path

Use these paper steps instead when your teacher chooses the no-fan path.

Goal: I can explain how wind can lift a load and use sample results to plan a fair test.

1. Draw and label the blades, hub, shaft, string and cup.
2. Read the practice table.
3. Explain what one result means in centimeters.

Keep: A labeled drawing and a correct reading of one result.

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Unit: Centimeters lifted in 10 seconds with one unchanged cup and load

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15